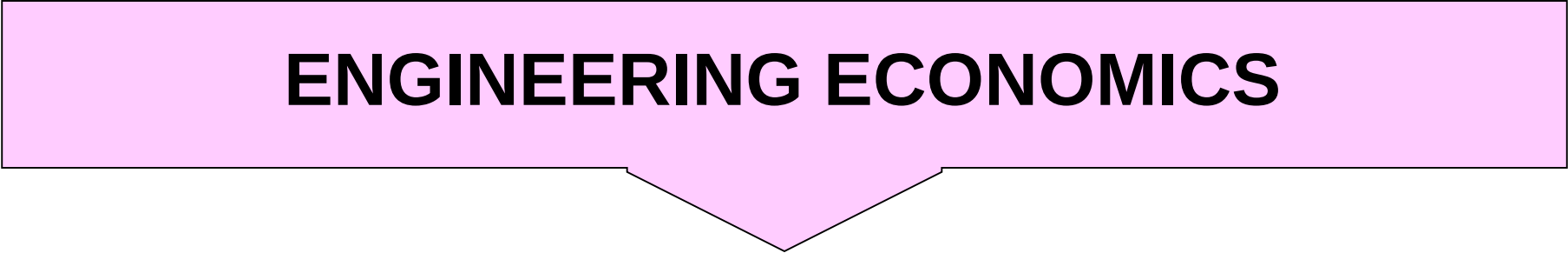
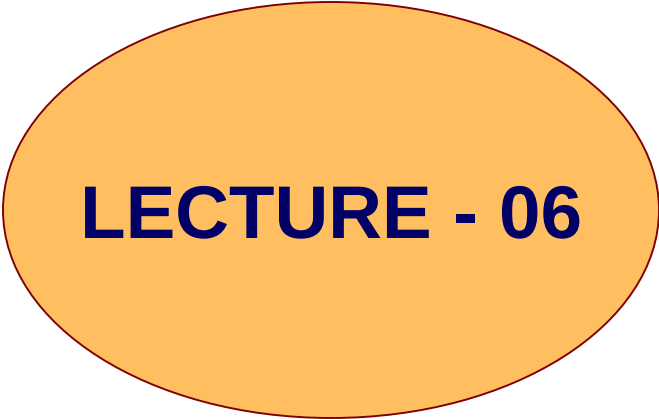


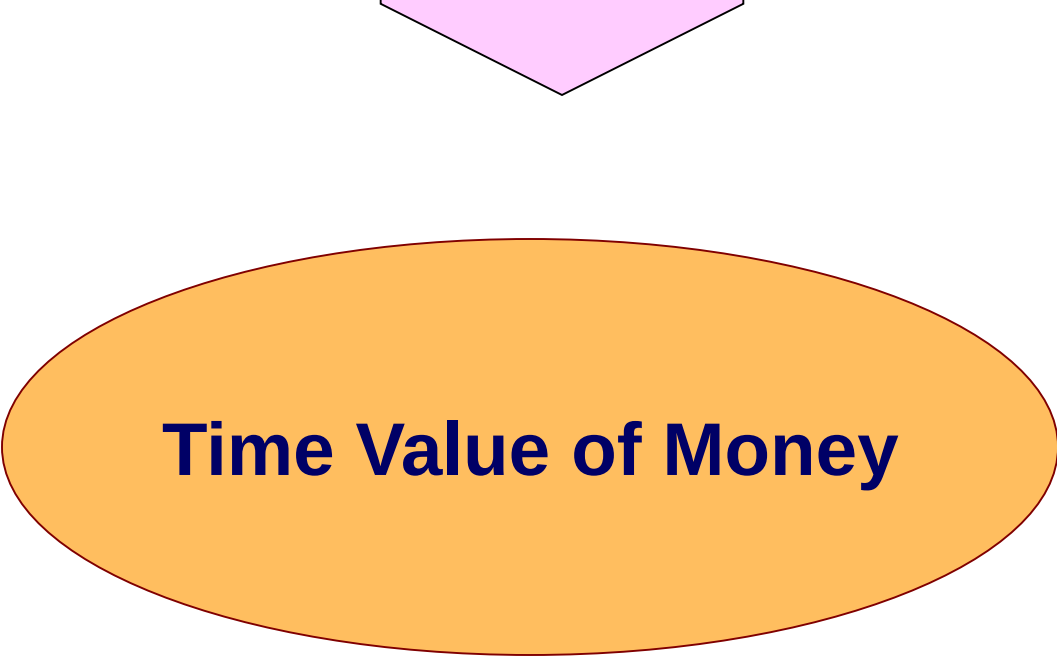
ENGINEERING ECONOMICS



LECTURE - 06



Opening Case



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graph TD; A[Opening Case] --> B([Time Value of Money]);
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Time Value of Money

Main Topics

- Earning Power of Money
- Purchasing Power of Money

Interest and Interest Rate

Interest

is the rental amount charged by financial institutions for the use of money.

Interest Rate

or the rate of capital growth, is the rate of gain received from an investment.

Usually this rate of gain is stated on a per year basis, and it represents the percentage gain realized on the money committed to the undertaking.

Thus, an 11% interest rate indicates that for every dollar of money used, an additional \$0.11 must be returned as payment for the use of money.

The interest rate is determined by mutual agreement between the borrower and the lender and is known as the **market rate**.

In one aspect, interest is an amount of money received as a result of investing funds either by lending it or by using it in the purchase of materials, labor or facilities. Interest received in this connection is gain or profit.

In another aspect, interest is an amount of money paid out as a result of borrowing funds. Interest paid in this connection is a cost.

The Time Value of Money

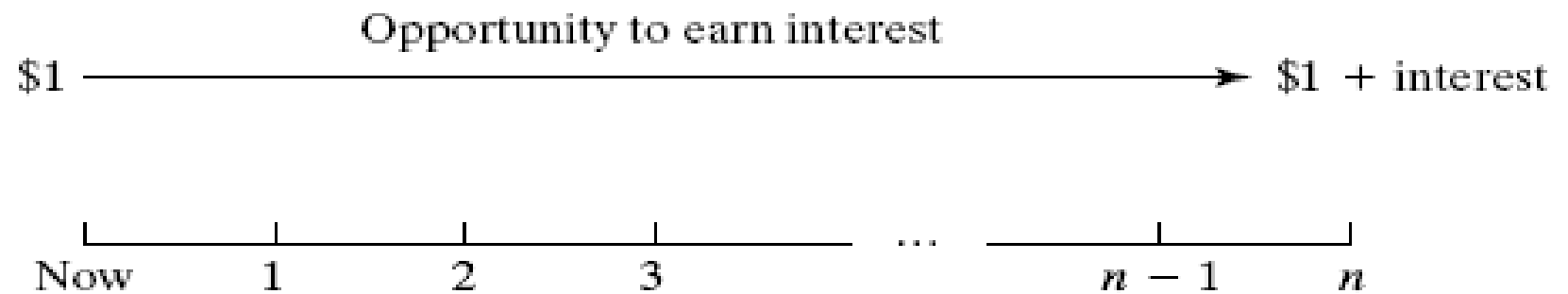
Because money can earn at a certain interest rate through its investment for a period of time, one thousand rupees received at some future date is not worth as much as in hand at present. This relationship between interest and time leads to the concept of the **time value of money**.

One thousand rupees in hand now is worth more than one thousand rupees received n years from now. Why?

Because having one thousand now provides the opportunity for investing that for n years more than the one thousand to be received at that time.

Since money has earning power, this opportunity will earn a return, so that after n years the original one thousand plus its interest will be greater than the one thousand received at that time. Thus, the fact that money has a time value means that equal amounts at different points in time have different value as long as the interest rate that can be earned exceeds zero.

This relationship between money and time is illustrated in fig on next slide



It is also true that money has time value because the purchasing power of a thousand changes through time.

During periods of inflation the amount of goods that can be bought for a particular amount of money decreases as the time of purchase occurs further out in the future.

Therefore, when considering the time value of money it is important to recognize both the earning power of money and the purchasing power of money.

The Earning Power of Money

Purchasing power is the value of a currency expressed in terms of the number of goods or services that one unit of money can buy. Purchasing power is important because, all else being equal, inflation decreases the number of goods or services you would be able to purchase.

The Earning Power of Money

Funds borrowed for the prospect of gain are commonly exchanged for goods, services, or instruments of production. This leads to the consideration of the **earning power of money** that may make it profitable to borrow.

Consider the example of Mr. ABC who manually digs ditches for underground cable. For this he is paid \$0.40 per foot and averages 200 feet per day. Weather conditions limit this kind of work to 180 days per year. Thus he has an income of \$80 per day worked or \$14,400 per year.

An advertisement brings to his attention a power ditcher that can be purchased for \$8000. Mr. ABC buys the ditcher after borrowing \$8000 at 14% interest. The machine will dig an average of 800 feet per day. By reducing the price to \$0.30 per foot he can get sufficient work to keep the machine busy when the weather will permit.

Estimated operating and maintenance costs for the ditching machine are \$40 per working day. At the end of year the machine is worthless because it is worn out. A summary of the venture follows:

- Receipts

• Amount of Loan	\$8000
• Payment for ditches dug	<u>43200</u>
	51200

- Disbursements

• Purchase of Ditcher	\$8000
• Operating and maintenance	7200
• Interest on loan	1120
• Repayment of loan	<u>8000</u>
	24320

- Receipts less Disbursements \$26880

An increase in net earnings for the year over the previous year of \$26880-\$14400=\$12480 is enjoyed by Mr. ABC.

This example is an illustration of what is known as the “earning power of money.”

It was an instrument of production, the power ditcher, that enables Mr. ABC to increase his earnings. Borrowed money made it possible for the instrument of production to be employed.

The Purchasing Power of Money

Purchasing power is the value of a currency expressed in terms of the number of goods or services that one unit of money can buy. Purchasing power is important because, all else being equal, inflation decreases the number of goods or services you would be able to purchase.

The Purchasing Power of Money

As prices increase or decrease, the amount of goods or services that can be purchased for a fixed amount of money decreases or increases accordingly.

Prices for goods and services are driven upward or downward because of numerous factors at work within the economy.

For example, increases in productivity and in the availability of goods tend to reduce prices, while government policies tend to increase prices. When all such effects are taken together, the most common result has been that prices increase.

Discussion